TaleCraft — Framework for 2D Point-and-Click Adventure Games

Alžbeta Kulichová | Supervisor: Mgr. Pavel Ježek, Ph.D.

Introduction

Despite its seemingly straightforward gameplay, creating a point-and-click adventure game involves implementing a variety of systems. For inexperienced developers, designing these can be difficult. Furthermore, there is a lack of accessible, beginner-friendly tools dedicated specifically to this genre. To fill this demand, the goal of this thesis is to design and implement an accessible framework tailored to the development of 2D point-and-click adventure games. This framework, called *TaleCraft*, will prioritize functionality and user-friendliness to help developers create games without need for advanced programming skills.

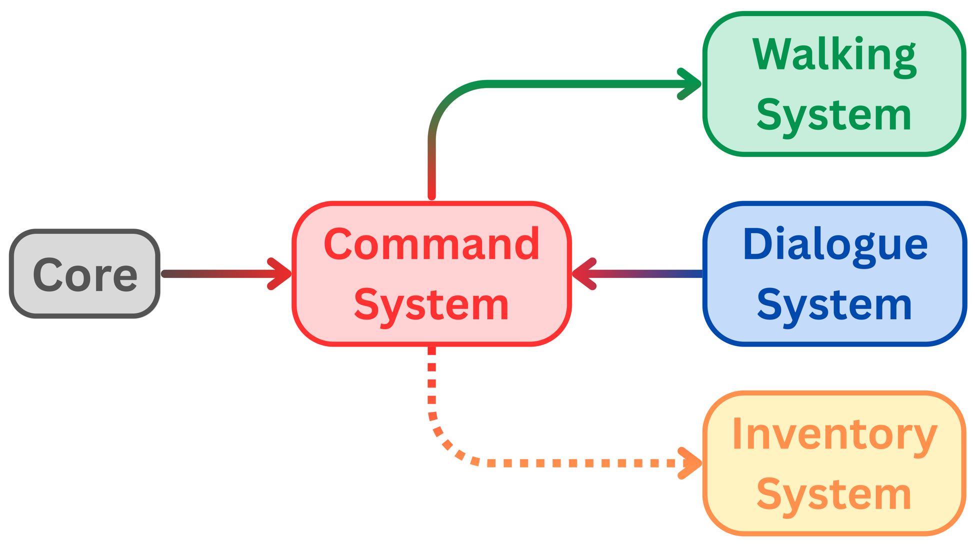
Goals

TaleCraft should meet the following goals:

- **Framework.** We implement the following systems:
  - Inventory
  - Commands
  - Movement
  - Dialogue
- **Demo games.** To showcase the utility of our framework, we create two demo games.

Architecture

The framework consists of five systems, four of them corresponding to the goals we aim to achieve. Fifth named Core manages key functionalities of the framework.



Walking System

Normally, the player has the ability to move around the walking area. TaleCraft supports this functionality by representing walkable areas using polygons. At least one polygon is required to define the main walkable region, while additional optional polygons can be used to define obstacles that characters must navigate around. Pathfinding is handled by an implementation of the A* algorithm. The system also supports character scaling based on their position, which creates the illusion of depth and makes the 2D environment appear more three-dimensional.

Command System

Typical point-and-click games use either a command-based or a context-based system to let the player interact with the world. To enable the use of both of these methods, we implemented a system that lets the user of the framework add new commands. If necessary, these commands can then be displayed on the screen and let them be selectable by the player. The system also offers additional features, such as the ability to show a sentence describing the actions that the player is about to take.

Dialogue

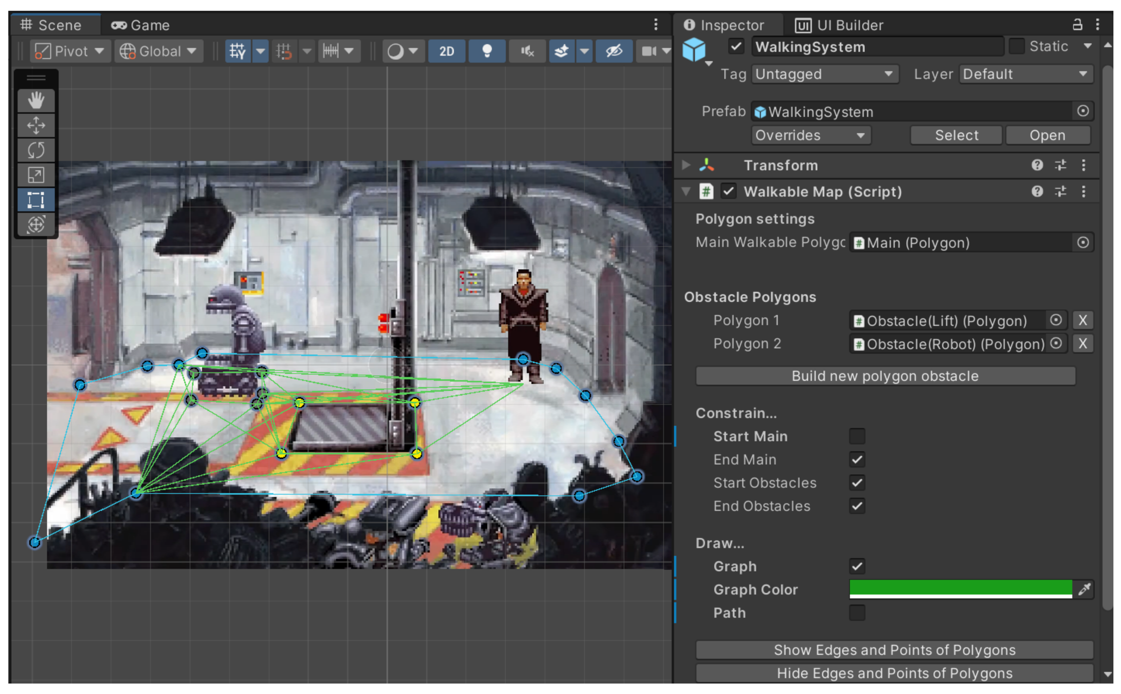
TaleCraft uses graphs for representing graphs. They are ideal when dealing with complex branching conversations. They also make it straightforward to implement cyclic paths, allowing characters to revisit or repeat parts of a conversation. In TaleCraft, the dialogue system is built using Unity's *GraphView* framework. Several types of nodes are used to construct these graphs:

- Start
- End
- Dialogue
- Choice
- Event
- Branch

Inventory

To let the player collect and use items, the framework includes an inventory system. A key feature is the ability to reference inventory items across different scenes. In TaleCraft, these items are `ScriptableObjects`, which makes it easy to create new items in the project and ensures they can be accessed globally. These objects store essential data such as the item's name and description, which can be used for features like displaying tags when the player hovers over an item with the mouse cursor.

Result: Framework



Result: Demo

Conclusion

The resulting framework delivers on its objectives and serves as a foundation for developing 2D point-and-click adventure games. In addition, two simple demo games that demonstrate the capabilities of our framework have been developed. Suggestions for future work include the implementation of features that should eventually be part of the framework, the most important being animation, sound, and saving system.

Info

Contact:
kulichova.alzbeta@gmail.com

Project GitHub Page:
https://github.com/bethkux/TaleCraft